

Query

If a platoon of soldiers is divided into brigades of 7 people each, 2 soldiers will not be included in any brigade. If the platoon is divided into brigades of 12 people each, again 2 soldiers will not be included in any brigade. What is the minimum number of soldiers that need to be added to the platoon so that it can be entirely divided into brigades of 7 people and brigades of 12 people?

Reasoning Summary

****Analyzing platoon size****

I'm working through a problem about a platoon size N where N is congruent to $2 \pmod{7}$ and 12 . Since both constraints yield the same remainder, N can also be written as congruent to $2 \pmod{84}$, the least common multiple of 7 and 12 . To make the platoon divisible by both brigades of 7 and 12 , I need to determine a value x that when added to N gives a multiple of 84 . That means x needs to equal 82 to achieve this.

****Clarifying platoon division****

Hmm, I'm considering whether the user wants the platoon to be divisible by both 7 and 12 , or just separately. To ensure $N+x$ is a multiple of both, I confirm that x must equal 82 since it brings us to the nearest multiple of 84 . However, if they want both options available at the same time, I need to clarify that reasoning. I'll summarize the steps clearly: recognizing congruences, using the least common multiple, and noting the minimal addition needed. It's essential to communicate this clearly!

****Evaluating additions for divisibility****

I'm double-checking my approach. Since N is congruent to $2 \pmod{7}$, adding 5 makes it divisible by 7 , while adding 10 does the same for 12 . But I need the same x for both equations. So I'll start by setting up x congruent to $5 \pmod{7}$ and $10 \pmod{12}$. To solve it, I can use the Chinese Remainder Theorem. I'll represent x as $7a + 5$ and find the suitable value of a to satisfy both equations. It's a good puzzle to crack!

****Confirming minimal addition****

I've represented x in terms of a to find out how many soldiers to add. By letting $a = 12t + 11$, I calculate x as 7 multiplied by that, resulting in $x = 84t + 82$. The minimal addition needed is clearly 82 . I want to present my conclusions simply and concisely, so I'll directly state that 82 soldiers need to be added and provide a brief explanation of the reasoning behind this. Time to finalize the answer!

Final Answer

Let the platoon size be N . We have $N \equiv 2 \pmod{7}$ and $N \equiv 2 \pmod{12}$. Since $\gcd(7,12)=1$, this implies $N \equiv 2 \pmod{84}$, so $N = 84k + 2$. To be divisible by both 7 and 12 , $N + x$ must be a multiple of 84 . Since $N \equiv 2 \pmod{84}$, we need $x \equiv -2 \equiv 82 \pmod{84}$. The smallest nonnegative x is 82 . So 82 soldiers must be added.

Zero-shot, Prompt-based Inversion

**** Analyzing platoon size ****

Alright, I'm diving into the problem where the platoon size (N) leaves a remainder of 2 when divided by both 7 and 12 . This means ($N \equiv 2 \pmod{7}$) and ($N \equiv 2 \pmod{12}$). Since 7 and 12 are coprime (their greatest common divisor is 1), the Chinese Remainder Theorem tells us that there is a unique solution modulo the product of 7 and 12 , which is 84 . Therefore, ($N \equiv 2 \pmod{84}$). This means (N) can be written as ($N = 84k + 2$) for some integer (k).

Now, the question is, how many soldiers need to be added to (N) so that the new total is divisible by both 7 and 12 ? In other words, I need to find the smallest nonnegative integer (x) such that ($N + x$) is a multiple of 84 . Since ($N = 84k + 2$), we need ($84k + 2 + x$) to be a multiple of 84 . This simplifies to ($x \equiv -2 \pmod{84}$). The smallest nonnegative (x) that satisfies this is 82 .

**** Clarifying platoon division ****

I need to ensure that the platoon can be divided into brigades of 7 and 12 soldiers each. The current platoon size (N) is $84k + 2$, which means it's 2 more than a multiple of 84 . To make it a multiple of 84 , I need to add 82 soldiers. This ensures that the new total is a multiple of 84 , and thus can be divided evenly into brigades of 7 and 12 soldiers.

**** Evaluating additions for divisibility ****

Let's break it down step by step. If ($N = 84k + 2$), then adding 5 to (N) makes it divisible by 7 , and adding 10 makes it divisible by 12 . However, I need a single value (x) that works for both conditions. I can use the Chinese Remainder Theorem to find such an (x). I represent (x) as ($x = 7a + 5$) and ($x = 12b + 10$). Setting these equal gives ($7a + 5 = 12b + 10$), or ($7a - 12b = 5$). I need to find integer solutions for (a) and (b). By trial and error or using the Extended Euclidean Algorithm, I find that ($a = 12t + 11$) and ($b = 7t + 5$) for some integer (t). Substituting ($a = 12t + 11$) into ($x = 7a + 5$) gives ($x = 7(12t + 11) + 5 = 84t + 82$). The smallest nonnegative (x) is 82 .

**** Confirming minimal addition ****

I've confirmed that the minimal addition needed is 82 . This is because ($N + 82 = 84k + 2 + 82 = 84k + 84 = 84(k + 1)$), which is a multiple of 84 . Therefore, the platoon can be divided into brigades of 7 and 12 soldiers each. The answer is clear and concise: 82 soldiers must be added to the platoon. So, the final answer is 82 soldiers.

Reasoning Trace Synthesized by Our Inversion Model

Okay, let's try to figure out this problem. So, there's a platoon of soldiers. When they divide them into brigades of 7 each, 2 soldiers are left out. Similarly, when they divide into brigades of 12 each, again 2 soldiers are left out. The question is asking for the minimum number of soldiers that need to be added so that the platoon can be divided evenly into brigades of both 7 and 12 without any leftovers.

First, let me parse the problem again to make sure I understand it correctly. The current size of the platoon, let's call it N , satisfies two conditions:

- When divided by 7 , the remainder is 2 .
- In mathematical terms, $N \equiv 2 \pmod{7}$.
- When divided by 12 , the remainder is also 2 . So, $N \equiv 2 \pmod{12}$.

We need to find the smallest number x such that $N + x$ is divisible by both 7 and 12 . Because if a number is divisible by both 7 and 12 , then it's divisible by their least common multiple (LCM). The LCM of 7 and 12 is 84 since they are coprime (their GCD is 1). Therefore, $N + x$ must be a multiple of 84 . So, our goal is to find the smallest x where $N + x \equiv 0 \pmod{84}$.

But first, we need to determine what N is. Wait, but the problem doesn't give us the exact number of soldiers in the platoon. It only tells us that $N \equiv 2 \pmod{7}$ and $N \equiv 2 \pmod{12}$. Hmm, so maybe we can find N first? But actually, perhaps we don't need to know N exactly because we can express x in terms of N .

Wait, but if N is congruent to $2 \pmod{7}$ and modulo 12 , then N must be congruent to $2 \pmod{\text{least common multiple of } 7 \text{ and } 12}$. Since 7 and 12 are coprime, their LCM is 84 . Therefore, $N \equiv 2 \pmod{84}$. So possible values of N are numbers like $2, 86, 170$, etc. But since we're looking for the minimal x to add to N to make it divisible by 84 , maybe we can ignore the actual value of N and just work with the congruence.

Wait, but maybe not. Because even though $N \equiv 2 \pmod{84}$, depending on what N is, the required x to reach the next multiple of 84 would vary. For example, if N is 2 , then x would be $84 - 2 = 82$. If N is 86 , then x would be $84 - 86 = -2$, which isn't possible, so we take $84 - (-2) = 86$. Wait, no. Let me think again.

If $N \equiv 2 \pmod{84}$, then N can be written as $84k + 2$ for some integer $k \geq 0$. To make $N + x$ divisible by 84 , we need $N + x = 84m$ for some integer m . Therefore, $x = 84m - N = 84m - (84k + 2) = 84(m - k) - 2$. To minimize x , we need the smallest positive x . The minimal x occurs when $m - k = 1$, so $x = 84 \cdot 1 - 2 = 82$. But wait, if N is $84k + 2$, then the next multiple of 84 is $84(k+1)$. So the difference is $84(k+1) - (84k + 2) = 84 - 2 = 82$. Therefore, regardless of k , the minimal x needed is 82 . Is that right?

[... reasoning steps omitted for space ...]

So I think that's correct. But let me test with specific numbers. Let's say N is 2 . Then adding 82 soldiers gives 84 . 84 divided by 7 is 12 , no remainder. 84 divided by 12 is 7 , no remainder.

Good. Another example: $N=86$. 86 divided by 7 is $12 \cdot 7 = 84$, remainder 2 . 86 divided by 12 is $7 \cdot 12 = 84$, remainder 2 . So adding 82 soldiers gives $86 + 82 = 168$. 168 divided by 7 is 24 , no remainder. Divided by 12 is 14 , no remainder. Correct.

Another example: $N=170$. 170 divided by 7 is $24 \cdot 7 = 168$, remainder 2 . Divided by 12 is $14 \cdot 12 = 168$, remainder 2 . Adding 82 : $170 + 82 = 252$. $252 / 7 = 36$, $252 / 12 = 21$. Correct.

Therefore, regardless of the starting N , as long as it's $2 \pmod{84}$, adding 82 soldiers will make it a multiple of 84 , hence divisible by both 7 and 12 . Therefore, the minimal number of soldiers to add is 82 .

But wait, the problem says "the minimum number of soldiers that need to be added to the platoon so that it can be entirely divided into brigades of 7 people and brigades of 12 people." So, perhaps the answer is 82 . But let me check once more.

Alternatively, maybe the problem is interpreted as needing to divide the entire platoon into both 7 and 12 brigades, meaning that the total number must be a multiple of both 7 and 12 . Therefore, the smallest number that is a multiple of both 7 and 12 is 84 . So the platoon's size N must be adjusted to the next multiple of 84 . Since $N \equiv 2 \pmod{84}$, the next multiple is $84 + (84 - 2) = 168$. Therefore, 82 soldiers need to be added. Yes, that's the same conclusion.

Therefore, the answer is 82 .